

草稿

记

$$\phi_t = \phi(k_t), \quad \psi_t = \phi(q_t), \quad S_t = (W_t, z_t).$$

1) Performer

Update:

$$\text{Update}_{\text{perf}}(S_{t-1}, k_t, v_t) : \begin{cases} W_t = W_{t-1} + v_t \phi_t^\top, \\ z_t = z_{t-1} + \phi_t. \end{cases}$$

Output:

$$\text{Output}_{\text{perf}}(S_t, q_t) = \frac{W_t \psi_t}{z_t^\top \psi_t}.$$

2) Performer + delta rule

先定义中间状态

$$S_t^- = \text{Update}_{\text{perf}}(S_{t-1}, k_t, -\text{Output}_{\text{perf}}(S_{t-1}, k_t)).$$

因为

$$\text{Output}_{\text{perf}}(S_{t-1}, k_t) = \frac{W_{t-1} \phi_t}{z_{t-1}^\top \phi_t},$$

所以第一步展开为

$$S_t^- : \begin{cases} W_t^- = W_{t-1} - \frac{W_{t-1} \phi_t}{z_{t-1}^\top \phi_t} \phi_t^\top, \\ z_t^- = z_{t-1} + \phi_t. \end{cases}$$

第二步:

$$S_t = \text{Update}_{\text{perf}}(S_t^-, k_t, v_t),$$

即

$$\begin{cases} W_t = W_t^- + v_t \phi_t^\top, \\ z_t = z_t^- + \phi_t. \end{cases}$$

代入回去得到最终 Update:

$$\text{Update}_\Delta(S_{t-1}, k_t, v_t) : \begin{cases} W_t = W_{t-1} - \frac{W_{t-1} \phi_t}{z_{t-1}^\top \phi_t} \phi_t^\top + v_t \phi_t^\top, \\ z_t = z_{t-1} + 2\phi_t. \end{cases}$$

Output:

$$\text{Output}_{\Delta}(S_t, q_t) = \frac{W_t \psi_t}{z_t^{\top} \psi_t} = \frac{\left(W_{t-1} + \left(v_t - \frac{W_{t-1} \phi_t}{z_{t-1}^{\top} \phi_t} \right) \phi_t^{\top} \right) \psi_t}{(z_{t-1} + 2\phi_t)^{\top} \psi_t}.$$